

Slovakia

Antimonopoly Office

I. Economic Analysis Data

With growing big data, accessibility The Antimonopoly Office of the Slovak Republic (AMOSR) started to use STATA 18.4 and R/Rstudio for more advanced data analysis. Stata is a powerful statistical software that enables users to manage large datasets, produce statistical analysis and graphical visualizations of data. It is primarily used by researchers in the fields of economics, biomedicine, and political science to examine data patterns.

In addition to STATA, analysts at the AMOSR also plan to work with R/RStudio which is used in data analysis to import, access, transform, explore, plot, and model data, and for machine learning to make predictions on data.

At the AMOSR we plan to primarily manage and analyse large economic data such as business register micro firm-level data, large own-firm accounting datasets, public procurement data, state aid data at the level of the recipient (businesses) and individual-level labor market data to provide evidence-based support in investigations carried out by the AMOSR. We also plan to publish regular economic briefs and materials on the state of competition in the various sectors of economic activity.

II. Forensic Tools

During the inspection, the Office obtains evidence of illegal anti-competitive conduct by the undertakings using specific IT technology with elements of forensic analysis. Digital forensic activity preserves the credibility of obtained digital traces and evidence, which are often subject to a court review.

The use of special tools and procedures in the identification, acquisition/restoration, processing, and analysis of data/information stored in electronic form by the undertaking guarantees that the obtained evidence is clearly derived from the obtained digital information (at the time of acquisition) and has not been altered in any way (integrity). The Office carries out inspections, or forensic activity at the premises of undertaking. The choice of software and hardware by the

Office is also adapted to this method of data acquisition and processing and to financial possibilities, with the aim of processing as much data as possible within the set time.

Due to the growing complexity and advanced level of IT systems, which undertakings use not only to store data, but also to conceal illegal evidence, the Office is aware of the need to use adequate tools for forensic data analysis, taking into account the effective use of funds for their procurement.

Currently, the Office has 6 pieces of mobile stations, which are powerful enough to work with complex forensic analysis tools, including other accessories (mobile printers, writeblockers, card readers, reductions, external disks and others).

Forensic analysis of mobile phones is performed using the UFED device with the appropriate software equipment (Cellebrite Physical Analyzer).

The main software tools of the Office are: FTK Imager (a tool for automatic creation of guaranteed copies of data - image), OS Forensics (a complex analytical tool designed for forensic data processing), and KillDisk (a tool for guaranteed deletion of data - wipe).

The Office also emphasizes the continuous training of employees ensuring the Office's forensic activities.

In an effort to adapt to current trends in the field of communication and data processing, the Office continuously reviews the state of forensic equipment. Based on the analysis, the Office concluded to improve/supplement the Office's technical equipment with a more robust analytical tool—NUIX, which is also used by other European competition authorities and shows high efficiency in processing data of various formats and sources.

III. Website Traffic Monitoring

SimilarWeb is an online platform that provides insights into website traffic, competitive analysis, and digital marketing. The platform collects and evaluates information from a multitude of diverse sources, which it then publishes on its website in a processed form. Collaborating platforms that provide data for SimilarWeb include portals such as Google and Facebook. SimilarWeb states on its

website that it has information on more than 100 million pages from over 190 countries.

For the Office, the most crucial function of SimilarWeb is the website traffic monitoring service, which tracks page visitation. In this category of services, SimilarWeb offers information about the number of visits to a specific page, unique visitors, the duration of page views, and even details on whether a visitor viewed the page from a mobile or PC browser. This information is not freely available from any other source.

Digital markets pose a challenge for the office in accurately determining market shares, which is essential for the proper assessment of the situation in a specific digital market. In the past, as part of its investigation, the Office used SimilarWeb to compare the traffic and popularity of different websites with the same or similar focus in order to determine their market share in the digital market. A tool such as the SimilarWeb platform could be a means to improve the Office's analytical capabilities.

IV. Public Procurement Analysis Data

In October/November 2023, the Slovak Competition Authority had access to the Transparex information system (database) available for testing—TRANSPAREX—Najväčšia databáza ekonomických a obchodných informácií. Transparex is probably the largest database of Slovak and Czech companies which mainly provides information on all legal entities, proprietary and personnel connections and conflicts of interest, public procurement data, a database of new orders, references from contract clients, information from the Land Register, data of the Register of beneficial owners.

The biggest advantage of this database was that we could access comprehensive information in one place about proprietary and personal connections of companies, their economic and financial indicators, but also important warnings about risk events, their participation in public tenders or their success. This was very helpful in identifying possible collusion in the public procurement process. The information was clearly processed and working with it was not time-consuming.

The information in this database is publicly and freely accessible but not in one place, so searching for information through this database saves time which was probably the biggest advantage for us.

The Transparex database would have been very beneficial to our work but due to the price and our financial capabilities it was not possible to purchase it.

V. Artificial Intelligence

Given the particularity of competition law and competition policy, it would be appropriate to have a suitable AI-based tool that would enable case-handlers to effectively search, link, and apply data from their own database of competition decisions, soft law, and jurisprudence.

The Office therefore has focused on the possibility of introducing an AI tool in the form of a user-friendly chat to search for and evaluate the application practice on two levels. The basic level is the database of non-public (full-format) decisions of the Antimonopoly Office usable for AI goals and an additional level represents the database of public decisions and jurisprudence of the European Commission and the Court of Justice of the EU with the possibility of expansion using other databases of important NCA 's decisions.

During the first half of 2024, the office has been looking for a suitable supplier of such an IT tool.